Cyber security Report : Phishing Email Detection And Awareness

Task 2: Future interns (FUTUTR_CS_02)

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Introduction

This project focuses on detecting and documenting email phishing as it one of the social engineering techniques used by attackers , as cyber security analyst i am tasked to detect suspicions emails ,using email analysing tools like analysing the senders information to see if the sender is an attacker or not and then the is manual analysis like identifying the message of sender, this assignment is for educational purpose only nothing unethical about the report .

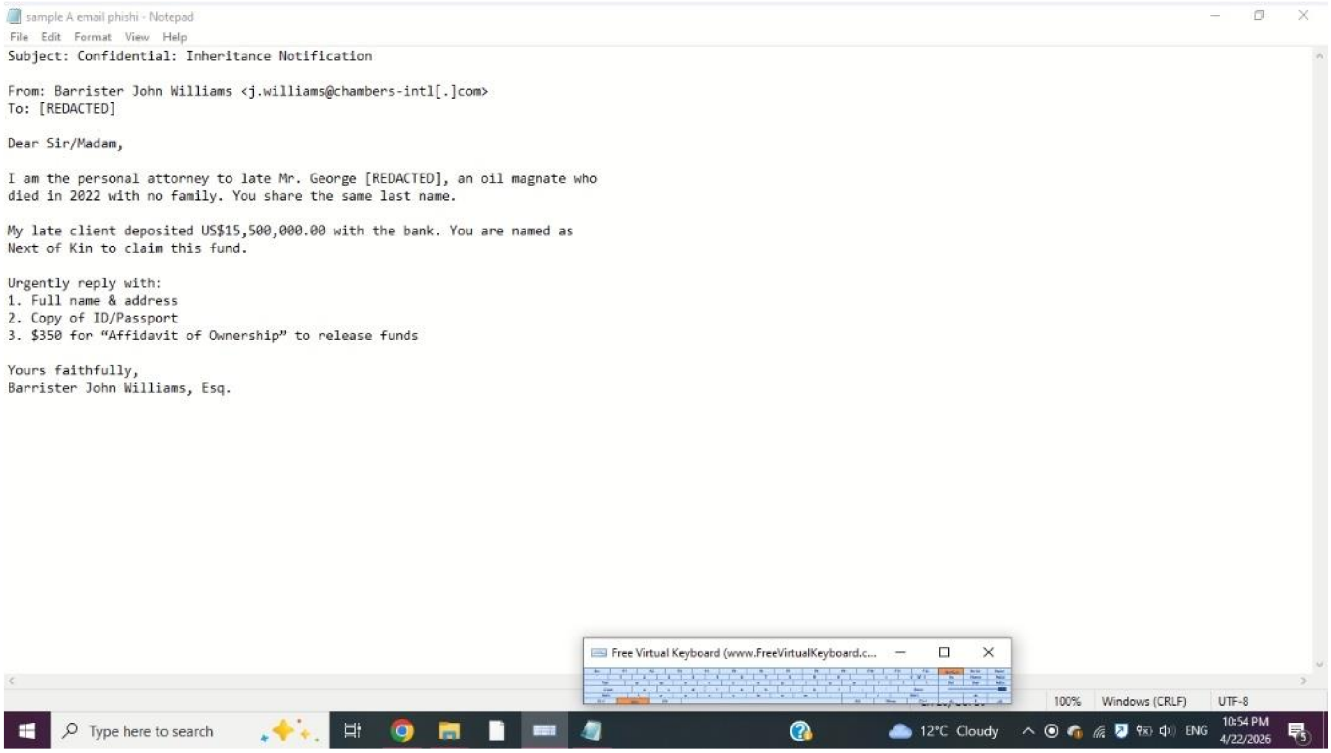
Tools Used

The following tools were used to analyze phishing emails:

- **Email Header Analysis Tools:**
  - Google Admin Toolbox (Message Header Analyzer)
  - MXToolbox Email Header Analyzer
- **Browser Safety Checks:**
  - Manual inspection of suspicious links (without clicking)
  - URL structure analysis (checking for fake domains and obfuscation)

Phishing email analysis

Sample A: Classic "Next of Kin" Inheritance



```
Subject: Confidential: Inheritance Notification

From: Barrister John Williams <j.williams@chambers-intl[.]com>
To: [REDACTED]

Dear Sir/Madam,

I am the personal attorney to late Mr. George [REDACTED], an oil magnate who
died in 2022 with no family. You share the same last name.

My late client deposited US$15,500,000.00 with the bank. You are named as
Next of Kin to claim this fund.

Urgently reply with:
1. Full name & address
2. Copy of ID/Passport
3. $350 for "Affidavit of Ownership" to release funds

Yours faithfully,
Barrister John Williams, Esq.
```

Summary of Attack

This email claims the recipient is entitled to a large inheritance from a deceased individual and requests personal information and a payment fee to process the claim.

Phishing Indicators

- **Unexpected message**: Recipient did not request or expect this email
- **Too good to be true**: Promise of $15.5 million inheritance
- **Request for sensitive data**: ID/passport and personal details
- **Advance fee scam**: Requests $350 to release funds
- **Generic greeting**: "Dear Sir/Madam" instead of using a name
- **Emotional manipulation**: Urgency and secrecy

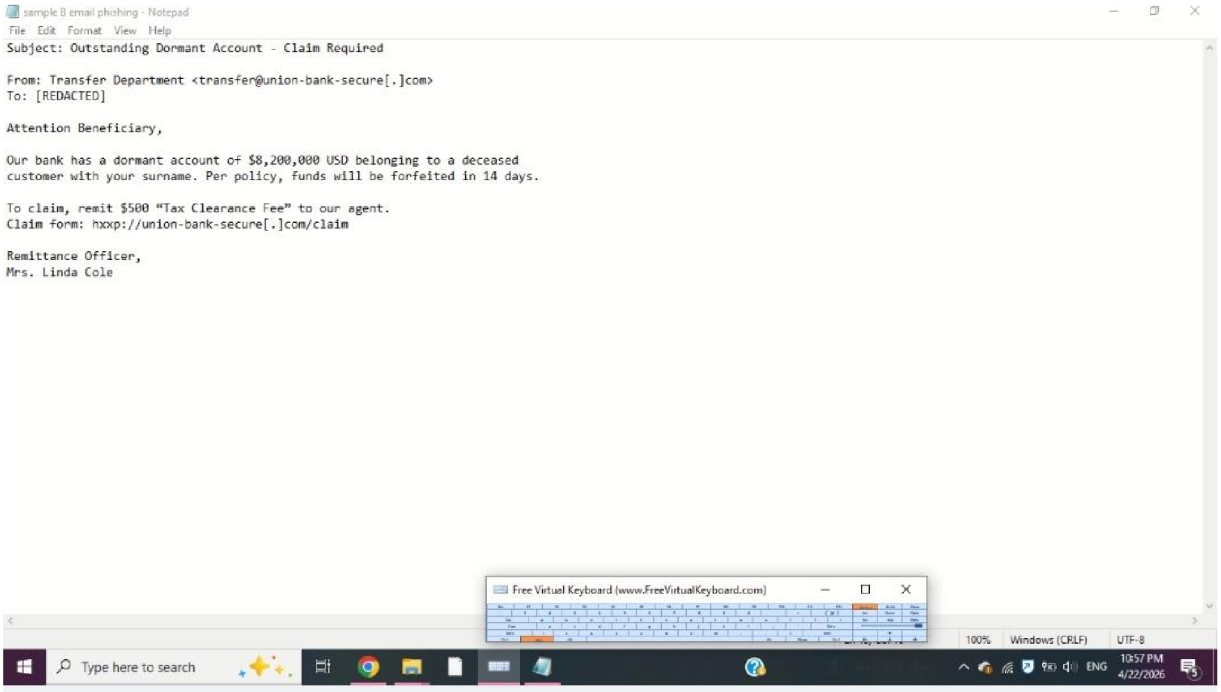
Risk Classification: HIGH RISK

This email is a classic **advance-fee fraud (419 scam)** designed to steal money and personal information.

Simple Explanation

The attacker is pretending to be a lawyer to trick you into believing you have inherited money. They ask for a small payment first, but once you pay, they may continue asking for more or disappear.

Section B : "Bank Notification" version



Summary of Attack

The email claims that a dormant bank account belonging to someone with your surname will be forfeited unless you act quickly and pay a fee.

Phishing Indicators

- **Fake urgency**: "Funds will be forfeited in 14 days"
- **Suspicious payment request**: $500 "Tax Clearance Fee"
- **Malicious link**: `hxxp://union-bank-secure[.]com/claim`
- **Impersonation of a bank**: Fake "Transfer Department"
- **Generic greeting**: "Attention Beneficiary" u
- **Unverified domain**: Not an official bank domain

Risk Classification: HIGH RISK

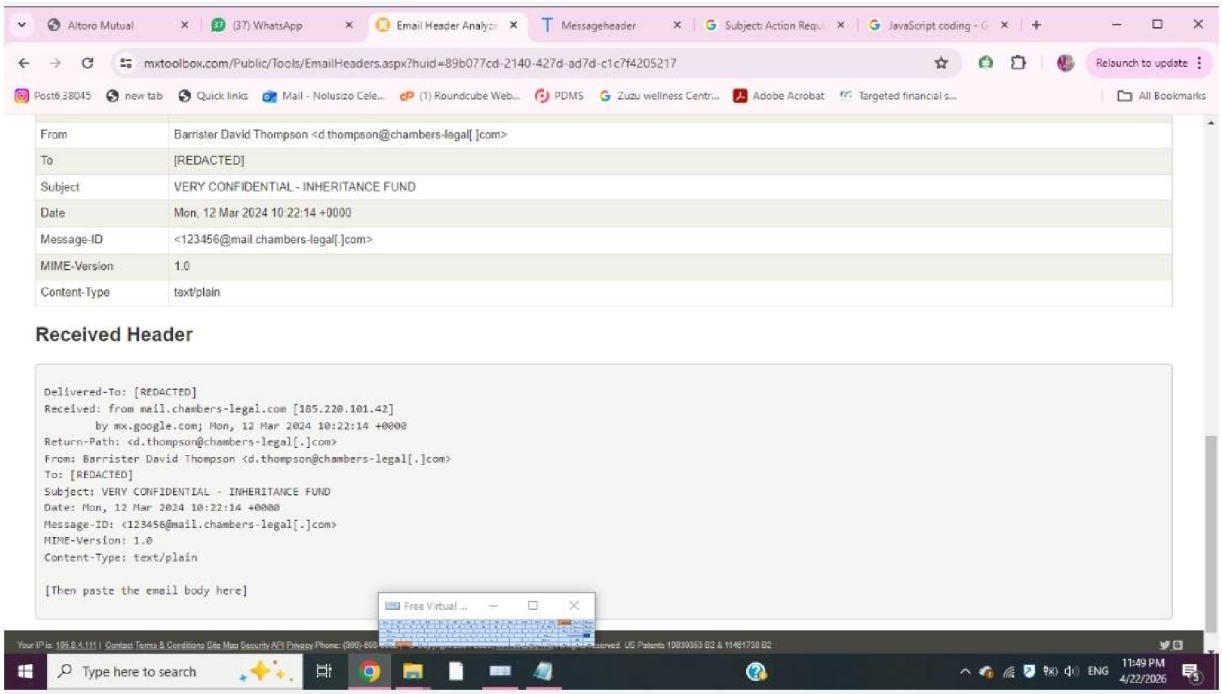
This is a **financial phishing scam** designed to:

- Steal money
- Direct victims to malicious websites
- Possibly collect login or personal data

Simple Explanation

The attacker is pretending to be a bank. They pressure you to act quickly and send money or click a link. The link may lead to a fake website designed to steal your information.

Sample C: Email Header Analysis



From	Barrister David Thompson <d.thompson@chambers-legal[.]com>
To	[REDACTED]
Subject	VERY CONFIDENTIAL - INHERITANCE FUND
Date	Mon, 12 Mar 2024 10:22:14 +0000
Message-ID	<123456@mail.chambers-legal[.]com>
MIME-Version	1.0
Content-Type	text/plain

Received Header

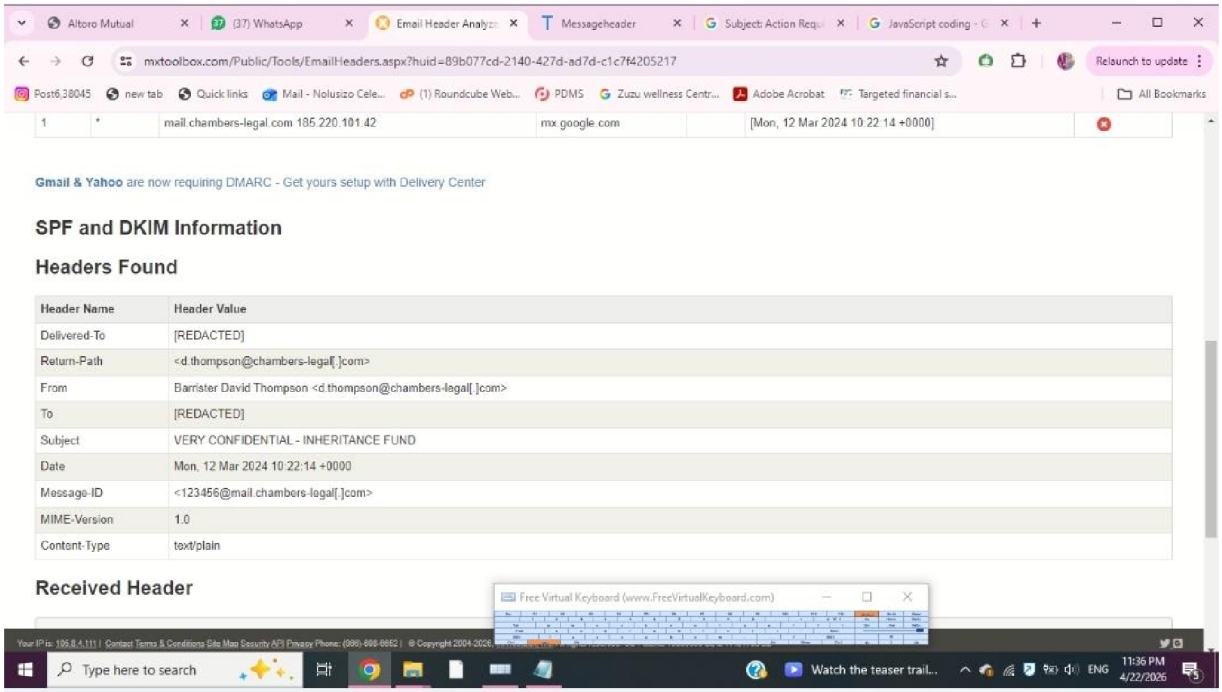
```
Delivered-To: [REDACTED]
Received: from mail.chambers-legal.com [185.220.101.42]
        by mx.google.com; Mon, 12 Mar 2024 10:22:14 +0000
Return-Path: <d.thompson@chambers-legal[.]com>
From: Barrister David Thompson <d.thompson@chambers-legal[.]com>
To: [REDACTED]
Subject: VERY CONFIDENTIAL - INHERITANCE FUND
Date: Mon, 12 Mar 2024 10:22:14 +0000
Message-ID: <123456@mail.chambers-legal[.]com>
MIME-Version: 1.0
Content-Type: text/plain

[Then paste the email body here]
```

Header Findings

- **Sending IP Address:** 185.220.101.42
- **Domain:** chambers-legal[.]com
- **Return Path mismatch:** May not match legitimate sender
- **Mail server origin:** Suspicious/unverified source

Phishing Indicators



- **Unknown sending server/IP**: Not linked to a trusted organization
- **Domain spoofing**: Looks professional but may be fake
- **Possible header inconsistencies**: Indicates spoofing
- **Lack of authentication**: (SPF/DKIM/DMARC missing or failing when checked in tools)

Risk Classification: HIGH RISK

The email likely originates from a **spoofed or malicious server**, indicating a phishing attempt.

Simple Explanation

Even though the email looks like it comes from a professional legal firm, the technical details show it was sent from an untrusted source. This is a strong sign of a fake email

Common Phishing Indicators Identified

Across all samples, the following warning signs were observed:

- **Unsolicited emails** (unexpected messages)
- **Large sums of money offered**
- **Requests for upfront payment**
- **Urgency or threats** ("act now or lose opportunity")
- **Generic greetings**
- **Suspicious links or domains**
- **Requests for sensitive information**
- **Impersonation of trusted entities (banks, lawyers)**

Risk Classification Summary

Sample	Type of attack	Risk level
A	Inheritance Scam	High
B	Bank phishing scam	High
C	Spoofed email headet	High

User Awareness Guide (Non-Technical)

How to Recognize Phishing Emails

- If it sounds **too good to be true**, it probably is
- Be cautious of emails asking for **money or personal details**
- Watch out for **urgent messages** pushing you to act quickly
- Check the sender's email address carefully
- Do not trust links without verifying them

How to Stay Safe

- **Do NOT click suspicious links**
- **Do NOT send money to unknown people**
- **Do NOT share personal or banking information via email**
- Verify messages by contacting the organization directly
- Use email security tools and spam filters

What to Do If You Receive a Phishing Email

- Do not reply
- Do not click links or download attachments
- Report the email as phishing
- Delete it immediately

Conclusion

Phishing attacks rely on deception, urgency, and trust manipulation. The analyzed samples demonstrate common tactics such as fake inheritance claims, impersonation of financial institutions, and spoofed email headers.

By understanding these warning signs and following basic safety practices, users can significantly reduce the risk of falling victim to phishing attacks.